

1. Project Description

AI FitTrack is a secure, scalable, AI-powered RESTful backend developed to simplify workout tracking and provide intelligent fitness recommendations. The application is built using Node.js, Express.js, MongoDB, and Mongoose.

The backend provides REST APIs for user authentication, workout management, workout searching, and AI-assisted fitness guidance. It follows a Model–View–Controller architecture with separate routes, controllers, services, middleware, models, and configuration modules.

Authentication uses JSON Web Tokens (JWT), while passwords are securely hashed using bcrypt.js before storage. Protected APIs validate JWT tokens through middleware before allowing access to personal resources.

Google Gemini AI is integrated through a dedicated service layer. It uses age, fitness goal, and experience level for personalized workout recommendations, and workout statistics such as total workouts, average duration, and calories burned for fitness insights.

Project Information

Project Name	AI-Augmented Backend Development AI FitTrack API
Team ID	6ab4cbbb0fc666a751b50cda
Date	29/09/2026
Backend	Node.js + Express.js
Database	MongoDB + Mongoose
AI	Google Gemini
Authentication	JWT + bcrypt.js

2. Scenario-Based Case Study

Background

Rahul is a fitness enthusiast who exercises regularly but records workouts manually using notebooks and spreadsheets. He finds it difficult to organize workout history, monitor progress, and determine whether his exercise routine matches his fitness goals. He also lacks personalized guidance for selecting suitable workouts.

Problem

- Difficulty maintaining a centralized workout record.
- Manual tracking can lead to incomplete workout history.
- No secure authentication for personal fitness records.
- Difficulty searching previous workouts efficiently.
- No personalized workout recommendations.
- No clear AI-generated insights about workout performance.

Solution

AI FitTrack provides a centralized backend where users can securely register, authenticate, and manage workout records through REST APIs. Authenticated users can create, retrieve, update, delete, and search workouts. Google Gemini AI provides intelligent workout recommendations and personalized fitness insights.

Usage

- Register an account.
- Login using email and password and receive a JWT token.
- Use the token to access protected APIs.
- Create, view, retrieve, update, and delete workout records.

- Search workouts by workout name, category, or workout date.
- Request AI workout recommendations and fitness insights.

3. Technical Architecture

AI FitTrack follows a layered MVC architecture. Routing, business logic, database interaction, authentication middleware, and AI services are separated into independent modules.

Request Flow

Client → Express Server → Routes → JWT Middleware → Controllers → Services / Models → MongoDB or Google Gemini → JSON Response

Architecture Components

- Client/API Consumer – React, mobile application, or Postman.
- Express Server – HTTP gateway, JSON parsing, CORS, routes, and errors.
- Authentication Middleware – validates JWT and identifies the user.
- Routes – map API endpoints to controller methods.
- Controllers – process requests and return standardized JSON.
- Services – reusable JWT, password, and Gemini AI logic.
- Models – User and Workout Mongoose schemas.
- MongoDB – persistent storage for users and workouts.
- Google Gemini Service – AI recommendation and insight generation.

API View Layer

Because AI FitTrack is a backend REST API, the View Layer is represented by API consumers such as web applications, mobile applications, and Postman. These clients send HTTP requests and receive JSON responses.

4. Pre-requisites

Software Requirements

- Windows 10/11, macOS, or Linux.
- Node.js v16 or above.
- npm v8 or above.
- Express.js.
- MongoDB.
- Postman or Thunder Client.
- Visual Studio Code or similar editor.
- Google Gemini API access for AI features.

Hardware Requirements

- Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.
- 8 GB RAM minimum; 16 GB recommended.
- At least 1 GB available storage.

Development Knowledge

- JavaScript and Node.js basics.
- Express.js REST API development.
- MongoDB and Mongoose.

- JWT authentication and bcrypt password hashing.
- API testing with Postman.
- Basic generative-AI API integration.

5. Project Workflow

The workflow below follows the milestone style of the sample documentation and is adapted to the actual FitTrack project.

Milestone 1: Architecture and Technology Selection

- Define the RESTful backend architecture.
- Select Node.js and Express.js.
- Select MongoDB and Mongoose.
- Define MVC/layered architecture.
- Plan JWT and bcrypt security.
- Plan Google Gemini integration through a service layer.

Milestone 2: Project Setup and Configuration

- Create the AI-FitTrack-API project and server structure.
- Initialize the Node.js project with `npm init -y`.
- Install required backend dependencies.
- Configure environment variables.
- Configure MongoDB connection.

Milestone 3: Backend Development

- Implement server and routes.
- Implement registration, login, and profile APIs.
- Implement JWT middleware.
- Implement workout CRUD and search.
- Implement Gemini recommendation and insight services.
- Add centralized error handling.

Milestone 4: Database and AI Integration

- Create User and Workout schemas.
- Maintain User-to-Workout one-to-many relationship.
- Validate workout fields.
- Prepare Gemini prompts from user details.
- Return personalized AI responses.

Milestone 5: Testing and Execution

- Start backend with the configured `npm start` command.
- Test authentication APIs.
- Test workout APIs.
- Test AI APIs.
- Verify validation, authorization, errors, and database operations.

6. Development Environment and Setup

Project Structure

```
AI-FitTrack-API/<br/>server/<br/> config/<br/> controllers/<br/> middleware/<br/> models/<br/> routes/<br/> services/<br/> .env<br/> package.json<br/> server.js
```

Initialize and Install

```
npm init -y<br/>npm install express mongoose dotenv bcryptjs jsonwebtoken cors<br/>npm install --save-dev nodemon
```

Environment Variables

The project documentation specifies PORT, MONGO_URI, JWT_SECRET, and GEMINI_API_KEY as environment variables. Secret values should not be exposed in source code.

```
PORT=5000<br/>MONGO_URI=mongodb://localhost:27017/aifittrack<br/>JWT_SECRET=your_secret_key<br/>GEMINI_API_KEY=your_google_gemini_api_key
```

7. Backend Development

Backend development implements REST APIs that allow authenticated users to manage workouts and interact with Google Gemini AI.

Major Modules

- Server Configuration
- Database Configuration
- Authentication
- Workout Management
- AI Recommendation
- Middleware
- Error Handling

Controllers

- Authentication Controller – registration, login, profile.
- Workout Controller – create, retrieve, update, delete, search.
- AI Controller – send prompts to Gemini, receive AI responses, and return results.

Service Layer

- Gemini AI Service – prepares prompts and formats recommendations.
- JWT Service – creates tokens, configures expiry, and verifies identity.
- Password Service – hashes passwords and compares passwords during login.

8. Database Configuration

MongoDB is the primary NoSQL database. Mongoose provides schema definitions, validation, relationships, and CRUD support.

Collections

- Users Collection
- Workouts Collection

User Schema

Field	Type	Description
_id	ObjectId	Primary key
name	String	User name
email	String	Unique email

Field	Type	Description
password	String	Encrypted password
createdAt / updatedAt	Date	Timestamps

Validation includes mandatory name and password, unique email, bcrypt password hashing, timestamps, and JWT support.

Workout Schema

Field	Type	Description
_id	ObjectId	Primary key
user	ObjectId	Reference to User
workoutName	String	Workout name
category	String	Workout category
duration	Number	Duration in minutes
caloriesBurned	Number	Calories burned
workoutDate	Date	Workout date
createdAt / updatedAt	Date	Timestamps

A User has many Workouts; each Workout belongs to one authenticated User through the user ObjectId.

9. API Functionalities

Authentication APIs

- Register User
- Login User
- Get Profile

Workout APIs

- Add Workout
- Get All Workouts
- Search Workouts
- Get Workout by ID
- Update Workout
- Delete Workout

AI APIs

- Workout Recommendation
- Fitness Insights

Supported HTTP Methods

- GET – retrieve resources
- POST – create resources or request AI processing
- PUT – update resources
- DELETE – remove workout records

Protected routes require a valid JWT token. Responses are returned in JSON format.

Authentication Flow

Register → Login → JWT → Protected Request → JWT Middleware → Controller → JSON Response

10. Google Gemini AI Integration

The AI module communicates with Google Gemini through a dedicated service layer. This keeps AI logic separate from controllers and database operations.

Workout Recommendation Inputs

- Age
- Fitness Goal
- Experience Level

Recommendation Outputs

- Personalized workout plan
- Weekly exercise suggestions
- Beginner / Intermediate / Advanced recommendations
- Motivational guidance
- Safety tips

Fitness Insight Inputs

- Total Workouts
- Average Workout Duration
- Total Calories Burned

Insight Outputs

- Performance analysis
- Improvement suggestions
- Motivational advice
- Fitness progress summary

AI Processing Flow

```
Client → AI Route → AI Controller → Gemini Service → Prompt → Google Gemini → AI Response → JSON
```

The source documentation identifies Google Gemini but does not specify an exact Gemini model name; therefore no model name is invented here.

11. Security and Error Handling

JWT Authentication

- JWT is generated after successful login.
- Protected APIs validate the token through middleware.
- Authenticated user information is attached to the request.
- Unauthorized requests are rejected.

Password Security

- Passwords are hashed using bcrypt.js before storage.
- Plain-text passwords are not intended to be stored.
- Password comparison is performed during login.

Centralized Error Handling

- Invalid credentials

- Missing authentication token
- Validation errors
- Database errors
- Internal server errors

Additional Protection

- Input validation
- Route protection
- Environment variables for secrets
- Standardized JSON error responses
- Modular separation of responsibilities

12. Testing and Project Execution

API testing is performed using Postman or Thunder Client. Testing is organized into authentication, workout, and AI APIs.

Authentication Testing

- Register User
- Login User
- Get Profile
- Invalid credential handling
- Missing/invalid JWT handling

Workout Testing

- Add Workout
- Get All Workouts
- Get Workout by ID
- Search Workouts
- Update Workout
- Delete Workout
- Validation rules

AI Testing

- Workout Recommendation
- Fitness Insights
- Required input validation
- JSON response verification

Execution

Install dependencies, configure environment variables, ensure MongoDB is available, and start the backend using the npm start command configured by the project. The exact command should be checked against package.json/README before final submission.

13. Project Outcome and Future Scope

AI FitTrack provides a centralized backend for secure workout tracking and AI-assisted fitness guidance. It replaces manual workout record keeping with authenticated REST APIs and organized MongoDB storage.

The project supports secure authentication, workout CRUD operations, workout search, Google Gemini recommendations, and AI fitness insights. Its modular MVC architecture separates models, controllers, routes, middleware, services, and configuration.

Future Extensions

- Nutrition tracking
- BMI calculation
- Wearable device integration
- Workout reminders
- Fitness analytics dashboards

Conclusion

AI FitTrack demonstrates how a Node.js and Express.js backend can combine secure authentication, MongoDB, RESTful API design, and generative AI to create a practical fitness tracking service. The modular structure supports future expansion while keeping responsibilities separated.

14. Repository and Submission Information

Repository

<https://github.com/naveenkbca2024-collab/Tn-Skill-AI-Arguments-FitTrack.git>

Submission Details

Project Name	AI-Augmented Backend Development AI FitTrack API
Team ID	6ab4cbbb0fc666a751b50cda
Date	29/09/2026
Backend	Node.js + Express.js
Database	MongoDB + Mongoose
AI Provider	Google Gemini
Testing	Postman / Thunder Client
Hosted URL	Not provided in source documentation
Demo Credentials	Not provided
Demo Video	A video link appears in the original project documentation; verify access before submission

Note: No deployment URL, credentials, measured performance results, or exact Gemini model name has been invented where the source documentation did not provide it.

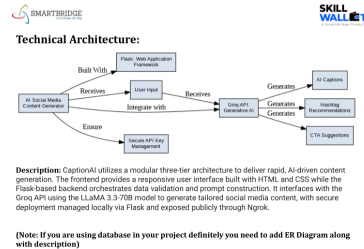
Reference Screenshots from the Uploaded Sample

The following screenshots are included as visual references from the uploaded CaptionAI sample documentation. They are not FitTrack application screenshots and are clearly marked for reference only.

Sample Documentation – Reference Screenshot 1

Sample Documentation – Reference Screenshot

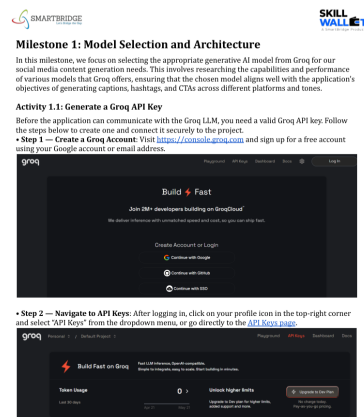
Visual reference from the uploaded CaptionAI sample documentation.



Sample Documentation – Reference Screenshot 2

Sample Documentation – Reference Screenshot

Visual reference from the uploaded CaptionAI sample documentation.



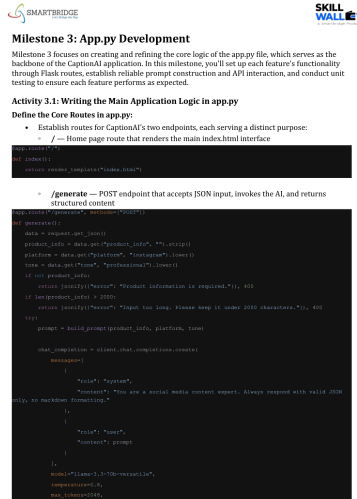
Reference Screenshots from the Uploaded Sample

Visual reference only — these images belong to the uploaded CaptionAI sample documentation.

Sample Documentation – Reference Screenshot 3

Sample Documentation – Reference Screenshot

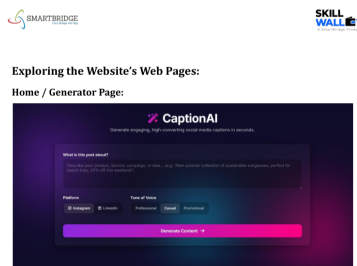
Visual reference from the uploaded CaptionAI sample documentation.



Sample Documentation – Reference Screenshot 4

Sample Documentation – Reference Screenshot

Visual reference from the uploaded CaptionAI sample documentation.



Description: The single-page interface of CaptionAI serves as both the landing page and the generation tool. It features a dark glassmorphism design with animated background blobs, a header displaying the CaptionAI brand (wand-magic-sparkles icon + logo), and a tagline. The main section contains the input form, platform toggle, tone selector, and the AI-generated results panel all on one seamless page.

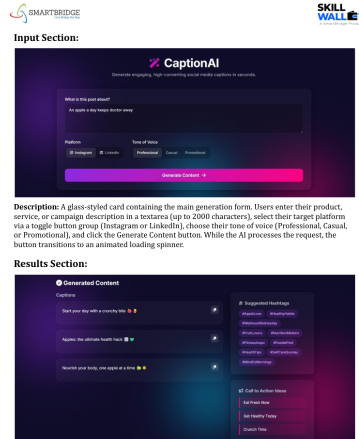
Reference Screenshots from the Uploaded Sample

Visual reference only — these images belong to the uploaded CaptionAI sample documentation.

Sample Documentation – Reference Screenshot 5

Sample Documentation – Reference Screenshot

Visual reference from the uploaded CaptionAI sample documentation.



Description: A glass-styled card containing the main generation form. Users enter their product, service, or campaign description in a textarea (up to 2000 characters), select their target platform via a toggle button group (Instagram or LinkedIn), choose their tone of voice (Professional, Casual, or Promotional), and click the Generate Content button. While the AI processes the request, the button transitions to an animated loading spinner.

Results Section:

Description: Revealed dynamically after generation, this section displays the AI output in a two-column grid. The left column shows three caption cards, each with the generated caption text and a copy-to-clipboard button. The right column contains two glass cards: one with ten hashtag chips (prefixed with #) and one with three CTA phrases in a list format. The viewport automatically smooth-scrolls to this section upon content generation.